

signals

Unix-like *operating system feature*

like exceptions, but for processes:

- can be triggered by an external process
 - kill command/system call

- can be triggered by special events
 - pressing control-C
 - other events that would normally terminate program
 - 'segmentation fault'
 - illegal instruction
 - divide by zero

- can invoke *signal handler* (like exception handler)

exceptions v signals

(hardware) exceptions	signals
handler runs in kernel mode	handler runs in user mode
hardware decides when	OS decides when
hardware needs to save PC	OS needs to save PC + registers
processor program counter changes	thread program counter changes
program counter = instruction to run next	

exceptions v signals

(hardware) exceptions	signals
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...but OS needs to run to trigger handler
most likely “forwarding” hardware exception

exceptions v signals

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hardware decides when	OS decides when
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signal handler follows normal calling convention
not special assembly like typical exception handler

exceptions v signals

(hardware) exceptions	signals
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hardware decides when	OS decides when
hardware needs to save PC	OS needs to save PC + registers
<i>processor</i> program counter changes	<i>thread</i> program counter changes
program counter = instruction to run next	

signal handler runs in same thread ('virtual processor')
as process was using before

not running at 'same time' as the code it interrupts

base program

```
int main() {  
    char buf[1024];  
    while (fgets(buf, sizeof buf, stdin)) {  
        printf("read_␣%s", buf);  
    }  
}
```

base program

```
int main() {  
    char buf[1024];  
    while (fgets(buf, sizeof buf, stdin)) {  
        printf("read %s", buf);  
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}
```

some input

read some input

more input

read more input

(control-C pressed)

(program terminates immediately)

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new program

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int main() {  
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read more input

(control-C pressed)

Control-C pressed?!

another input **read another input**

example signal program

```
void handle_sigint(int signum) {  
    /* signum == SIGINT */  
    write(1, "Control-C\ncpress?!\n", sizeof("Control-C\ncpress?!\n"));  
}  
  
int main(void) {  
    struct sigaction act;  
    act.sa_handler = &handle_sigint;  
    sigemptyset(&act.sa_mask);  
    // SA_RESTART = if syscall interrupted,  
    // complete it when handler returns  
    act.sa_flags = SA_RESTART;  
    sigaction(SIGINT, &act, NULL);  
  
    char buf[1024];  
    while (fgets(buf, sizeof buf, stdin)) {  
        printf("read\nc%s", buf);  
    }  
}
```

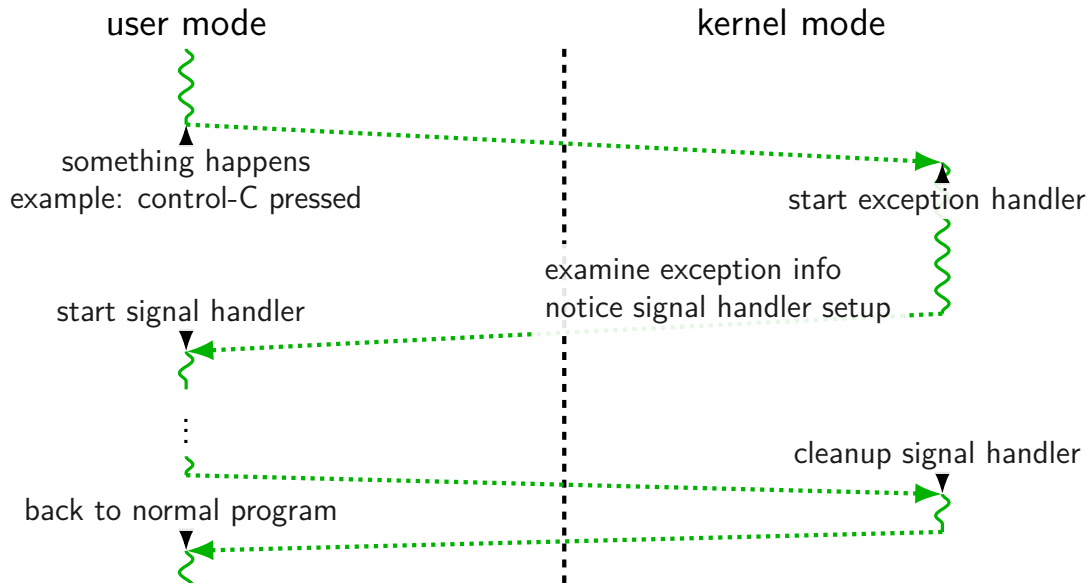
example signal program

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    act.sa_flags = SA_RESTART;  
    sigaction(SIGINT, &act, NULL);  
  
    char buf[1024];  
    while (fgets(buf, sizeof buf, stdin)) {  
        printf("read\nc%s", buf);  
    }  
}
```

'forwarding' exception as signal



SIGxxxx

signals types identified by number...

constants declared in `<signal.h>`

constant	likely use
SIGBUS	"bus error"; certain types of invalid memory accesses
SIGSEGV	"segmentation fault"; other types of invalid memory accesses
SIGINT	what control-C usually does
SIGFPE	"floating point exception"; includes integer divide-by-zero
SIGHUP, SIGPIPE	reading from/writing to disconnected terminal/socket
SIGUSR1, SIGUSR2	use for whatever you (app developer) wants
SIGKILL	terminates process (cannot be handled by process!)
SIGSTOP	suspends process (cannot be handled by process!)
...	...

SIGxxxx

signals types identified by number...

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SIGKILL	terminates process (<i>cannot be handled by process!</i>)
SIGSTOP	suspends process (<i>cannot be handled by process!</i>)
...	...

handling Segmentation Fault

```
...  
void handle_sigsegv(int num) {  
    puts("got SIGSEGV");  
}  
  
int main(void) {  
    struct sigaction act;  
    act.sa_handler = handle_sigsegv;  
    sigemptyset(&act.sa_mask);  
    act.sa_flags = SA_RESTART;  
    sigaction(SIGSEGV, &act, NULL);  
  
    asm("movq %rax, 0x12345678");  
}
```

handling Segmentation Fault

```
...  
void handle_sigsegv(int num) {  
    puts("got SIGSEGV");  
}  
  
int main(void) {  
    struct sigaction act;  
    act.sa_handler = handle_sigsegv;  
    sigemptyset(&act.sa_mask);  
    act.sa_flags = SA_RESTART;  
    sigaction(SIGSEGV, &act, NULL);  
  
    asm("movq %rax, 0x12345678");  
}
```

got SIGSEGV

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got SIGSEGV

signal API

`sigaction` — register handler for signal

`kill` — send signal to process

uses *process ID* (integer, retrieve from `getpid()`)

`pause` — put process to sleep until signal received

`sigprocmask` — temporarily block/unblock some signals from being received

signal will still be *pending*, received if unblocked

... and much more

kill command

kill command-line command : calls the `kill()` function

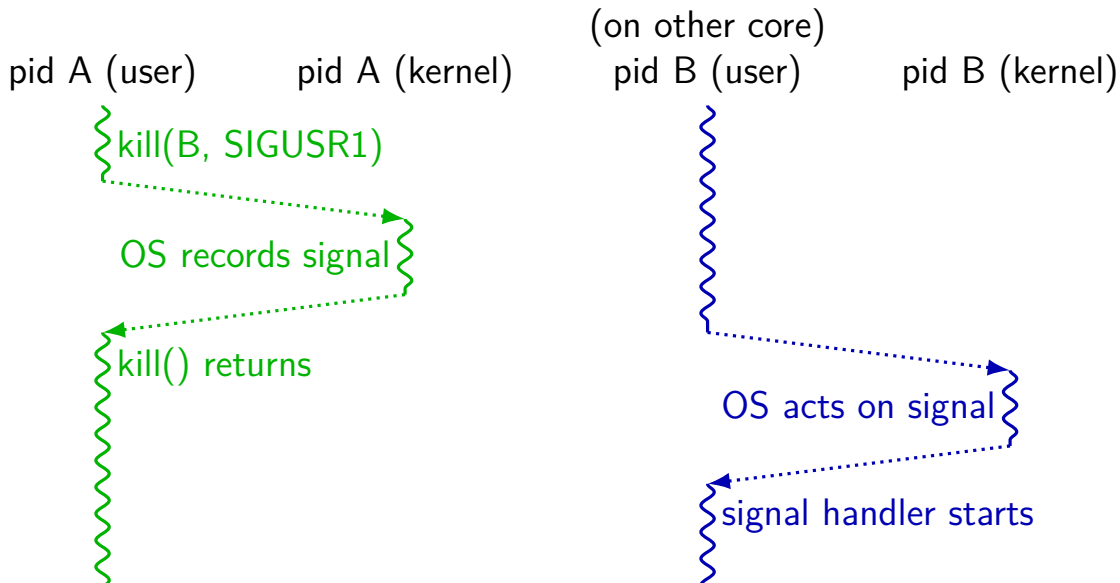
`kill 1234` — sends `SIGTERM` to pid 1234

in C: `kill(1234, SIGTERM)`

`kill -USR1 1234` — sends `SIGUSR1` to pid 1234

in C: `kill(1234, SIGUSR1)`

kill() not always immediate



output of this?

pid 1000

pid 2000



```
void handle_usr1(int num) {  
    write(1, "Y", 1);  
    _exit(0);  
}  
  
int main() {  
    struct sigaction act;  
    ... // initialize rest of "act"  
    act.sa_handler = &handle_usr1;  
    sigaction(SIGUSR1, &act, NULL);  
}
```

If these run at same time, expected output?

- A. XY
- B. X
- C. Y
- D. YX
- E. X or XY, depending on timing
- F. crash
- G. (nothing)
- H. something else

output of this? (v2)

pid 1000

pid 2000



```
void handle_usr1(int num) {
    write(1, "Y", 1);
    _exit(0);
}
int main() {
    struct sigaction act;
    ... // initialize rest of "act"
    act.sa_handler = &handle_usr1;
    sigaction(SIGUSR1, &act);
    while (1) pause(); // ADDED
}
```

If these run at same time, expected output?

- A. XY
- B. X
- C. Y
- D. YX
- E. X or XY, depending on timing
- F. crash
- G. (nothing)
- H. something else

signal handler unsafety (0)

```
void foo() {  
    /* SIGINT might happen while foo() is running */  
    char *p = malloc(1024);  
    ...  
}  
  
/* signal handler for SIGINT  
   (registered elsewhere with sigaction()) */  
void handle_sigint() {  
    printf("You pressed control-C.\n");  
}
```

signal handler unsafety (1)

```
void *malloc(size_t size) {  
    ...  
    to_return = next_to_return;  
    /* SIGNAL HAPPENS HERE */  
    next_to_return += size;  
    return to_return;  
}  
  
void foo() {  
    /* This malloc() call interrupted */  
    char *p = malloc(1024);  
    p[0] = 'x';  
}  
  
void handle_sigint() {  
    // printf might use malloc()  
    printf("You pressed control-C.\n");  
}
```

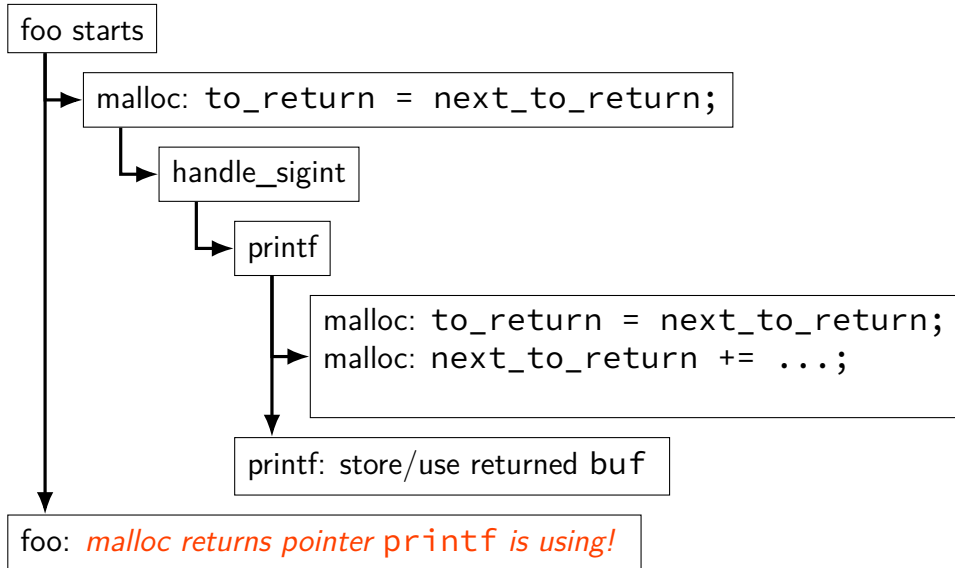
signal handler unsafety (1)

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void *malloc(size_t size) {  
    ...  
    to_return = next_to_return;  
    /* SIGNAL HAPPENS HERE */  
    next_to_return += size;  
    return to_return;  
}  
  
void foo() {  
    /* This malloc() call interrupted */  
    char *p = malloc(1024);  
    p[0] = 'x';  
}  
  
void handle_sigint() {  
    // printf might use malloc()  
    printf("You pressed control-C.\n");  
}
```

signal handler unsafety (2)

```
void handle_sigint() {  
    printf("You pressed control-C.\n");  
}  
  
int printf(...) {  
    static char *buf;  
    ...  
    buf = malloc()  
    ...  
}
```

signal handler unsafety: timeline



signal handler unsafety (3)

```
foo() {  
    char *p = malloc(1024)... {  
        to_return = next_to_return;  
        handle_sigint() { /* signal delivered here */  
            printf("You pressed control-C.\n") {  
                buf = malloc(...) {  
                    to_return = next_to_return;  
                    next_to_return += size;  
                    return to_return;  
                }  
                ...  
            }  
        }  
        next_to_return += size;  
        return to_return;  
    }  
    /* now p points to buf used by printf! */  
}
```

signal handler unsafety (3)

```
foo() {  
    char *p = malloc(1024)... {  
        to_return = next_to_return;  
        handle_sigint() { /* signal delivered here */  
            printf("You pressed control-C.\n") {  
                buf = malloc(...) {  
                    to_return = next_to_return;  
                    next_to_return += size;  
                    return to_return;  
                }  
                ...  
            }  
        }  
        next_to_return += size;  
        return to_return;  
    }  
    /* now p points to buf used by printf! */  
}
```


signal handler safety

POSIX (standard that Linux follows) defines “async-signal-safe” functions

`man signal-safety` can get you list on portal

these must work correctly no matter what they interrupt

...and no matter how they are interrupted

includes: `write`, `_exit`

does not include: `printf`, `malloc`, `exit`

blocking signals

avoid having signal handlers anywhere:

can instead *block signals*

`sigprocmask()`, `pthread_sigmask()`

blocked = signal handler doesn't run

also called “masking” a signal

blocked signals are not *delivered* (acted on)

instead, signal becomes *pending*

delivered if unblocked

blocking signals

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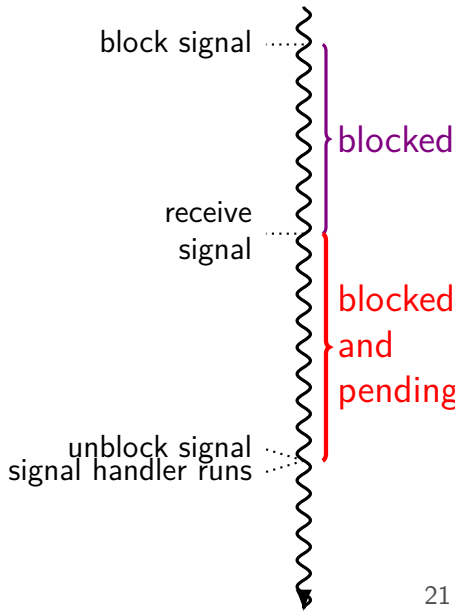
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controlling when signals are handled

first, block a signal

then either unblock signals only at certain times

some special functions to help:

`sigsuspend` (unblock and wait until handler runs),

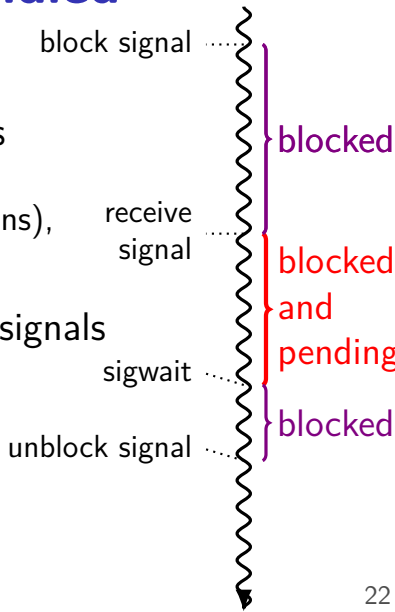
`pselect` (unblock while checking for I/O), ...

and/or use API for checking/changing pending signals

example: `sigwait`

(wait for blocked signal to become pending;
then mark not pending)

typically *instead of having signal handler*



controlling when signals are handled

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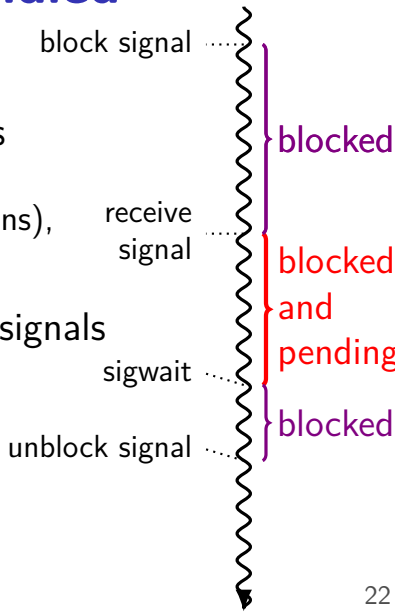
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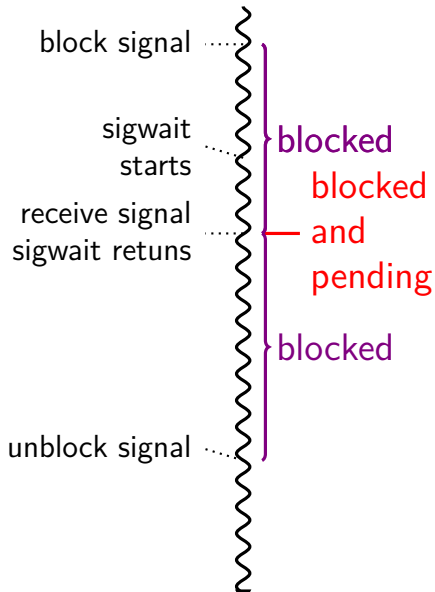
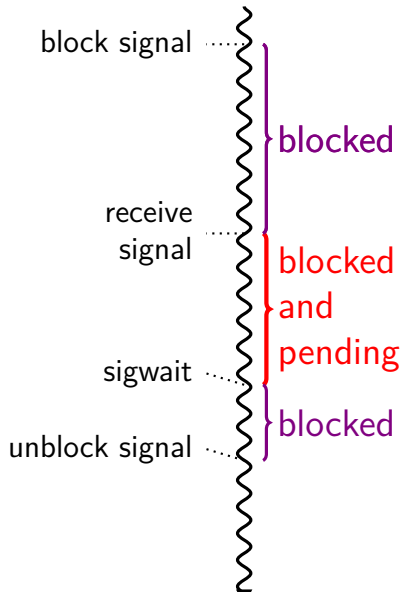
example: `sigwait`

(wait for blocked signal to become pending;
then mark not pending)

typically *instead of having signal handler*



sigwait timelines



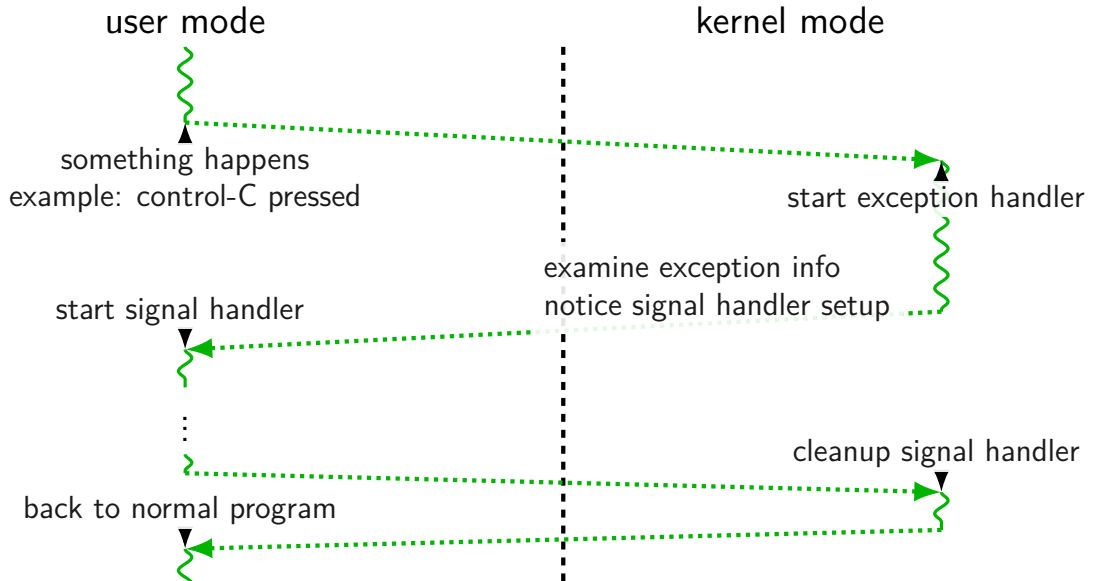
synchronous signal handling

```
int main(void) {
    sigset_t set;
    sigemptyset(&set);
    sigaddset(&set, SIGINT);
    sigprocmask(SIG_BLOCK, &set, NULL);

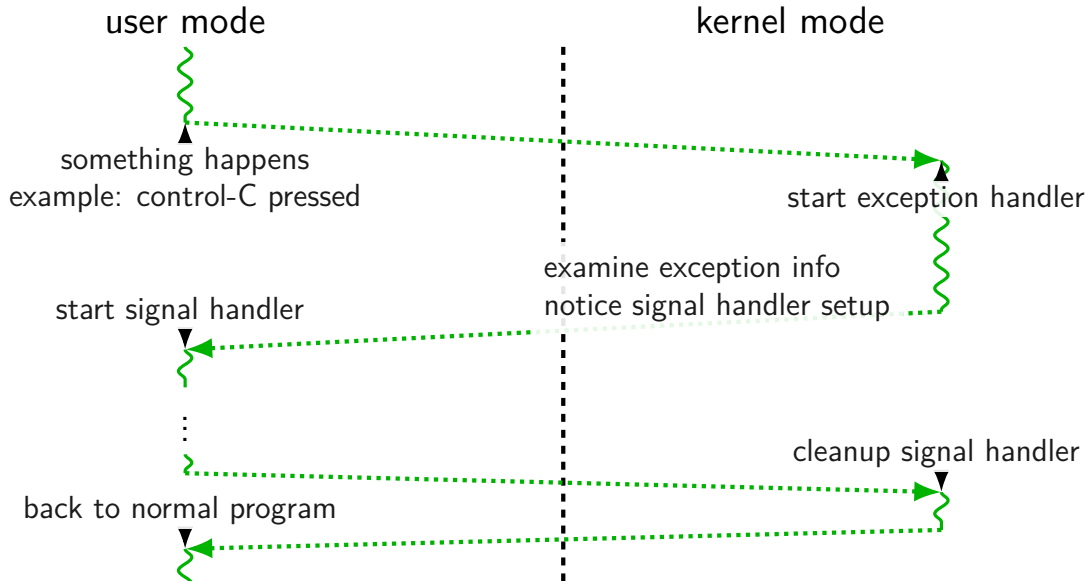
    printf("Waiting for SIGINT (control-C)\n");
    int num;
    if (sigwait(&set, &num) != 0) {
        printf("sigwait failed!\n");
    }
    if (num == SIGINT);
        printf("Got SIGINT\n");
    }
}
```

backup slides

'forwarding' exception as signal



'forwarding' exception as signal



x86-64 Linux signal delivery (1)

suppose: signal (with handler) happens while `foo()` is running

should stop in the middle of `foo()`

do signal handler

go back to `foo()` without...

changing local variables (possibly in registers)

(and `foo()` doesn't have code to do that)

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go back to `foo()` *without...*

changing local variables (possibly in registers)

(and `foo()` doesn't have code to do that)

x86-64 Linux signal delivery (2)

suppose: signal (with handler) happens while `foo()` is running

OS saves registers *to user stack*

OS modifies user registers, PC to call signal handler

the stack

address of <code>__restore_rt</code>
saved registers
PC when signal happened
local variables for <code>foo</code>
...

→ stack pointer
when signal handler started

→ stack pointer
before signal delivered

x86-64 Linux signal delivery (3)

```
handle_sigint:
```

```
...
```

```
ret
```

```
...
```

```
__restore_rt:
```

```
// 15 = "sigreturn" system call
```

```
movq $15, %rax
```

```
syscall
```

__restore_rt is *return address* for signal handler

sigreturn syscall restores pre-signal state

- if SA_RESTART set, restarts interrupted operation

- also handles caller-saved registers

- also might change which signals blocked (depending how sigaction was called)

SA_RESTART

```
struct sigaction sa; ...  
sa.sa_flags = SA_RESTART;
```

general version:

```
sa.sa_flags = SA_NAME | SA_NAME | SA_NAME; (or 0)
```

if SA_RESTART included:

after signal handler runs, attempt to restart interrupted operations (e.g. reading from keyboard)

if SA_RESTART not included:

after signal handler runs, interrupted operations return typically an error (detect by checking `errno == EINTR`)

sending signals (1)

PID 1000:

```
void handle_usr1(int num) {
    write(1, "Y", 1);
    kill(2000, SIGUSR2);
}

int main() {
    struct sigaction act;
    ... // initialize act
    act.sa_handler = &handle_usr1;
    sigaction(SIGUSR1, &act, NULL);
    // wait for pid 2000 to start
    sleep(60);
    kill(2000, SIGUSR1);
    while (1) pause();
}
```


sending signals (2)

PID 1000:

```
void handle_usr1(int num) {
    write(1, "Y", 1);
    kill(2000, SIGUSR2);
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int main() {
    struct sigaction act;
    ... // initialize act
    act.sa_handler = &handle_usr1;
    sigaction(SIGUSR1, &act, NULL);
    sleep(60); // wait for pid 2000
    kill(2000, SIGUSR1);
    while (1) pause();
}
```

PID 2000:

```
void handle_usr1(int num) {
    write(1, "X", 1);
    kill(1000, SIGUSR1);
}
```

```
void handle_usr2(int num) {
    write(1, "Z", 1);
    kill(1000, SIGTERM);
    _exit(0);
}
```

```
int main() {
    struct sigaction act;
    ... // initialize act
    act.sa_handler = &handle_usr1;
    sigaction(SIGUSR1, &act, NULL);
    act.sa_handler = &handle_usr2;
    sigaction(SIGUSR2, &act, NULL);
    while (1) pause();
}
```

sending signals (2)

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void handle_usr1(int num) {  
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void handle_usr2(int num) {  
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}
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```
int main() {  
    struct sigaction act;  
    act.sa_handler = &handle_usr1;  
    sigaction(SIGUSR1, &act, NULL);  
    act.sa_handler = &handle_usr2;  
    sigaction(SIGUSR2, &act, NULL);  
    while (1) pause();  
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```

sending signals (2)

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sending signals (2)

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    sigaction(SIGUSR1, &act, NULL);  
    sleep(60); // wait for pid 2000 to start  
    kill(2000, SIGUSR1);  
    while (1) pause();  
}
```

PID 2000:

```
void handle_usr1(int num) {  
    write(1, "X", 1);  
    kill(1000, SIGUSR1);  
}
```

```
void handle_usr2(int num) {  
    write(1, "Z", 1);  
    kill(1000, SIGTERM);  
    _exit(0);  
}
```

```
int main() {  
    struct sigaction act;  
    act.sa_handler = &handle_usr1;  
    sigaction(SIGUSR1, &act, NULL);  
    act.sa_handler = &handle_usr2;  
    sigaction(SIGUSR2, &act, NULL);  
    while (1) pause();  
}
```